

Cortical Connections of Striate and Extrastriate Visual Areas in Tree Shrews

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ABSTRACT

The ipsilateral and contralateral cortical connections of visual cortex of tree shrews (*Tupaia belangeri*) were investigated by placing restricted injections of fluorochrome tracers, wheat germ agglutinin-horseradish peroxidase, or biotinylated dextran amine into area 17 (V1), area 18 (V2), or the adjoining temporal dorsal area (TD). As previously reported, V1 was characterized by a widespread, patchy pattern of intrinsic connections; ipsilateral connections with V2, TD, and to a lesser extent, other areas of the temporal cortex; and contralateral connections with V1, V2, and TD. A surface-view of the myelin pattern in V1 revealed a patchwork of light and dark module-like regions. The ipsilateral connections with V2 and TD were roughly topographic, whereas heterotopic locations in V1 were callosally connected. Injections in V2 labeled as much as one third of V2 in a patchy pattern, and portions of ipsilateral V1 and TD in roughly topographic patterns. In addition, connections with several other visual areas in the temporal lobe were revealed. Contralaterally, most of the label was in V2, with some in V1 and TD. Injections in TD demonstrated connections within the region, and with adjoining portions of the temporal cortex, V2, and V1. There were sparse connections with an oval of densely myelinated cortex, which we have termed the temporal inferior area (TI). Callosal connections were concentrated in TD, but also included V2. The results provide further evidence for modular organizations within V1 and V2, and reveal for the first time the complete patterns of cortical connections of V2 and TD. The results are consistent with the proposal that at least three visual areas, the temporal anterior area, TA, the temporal dorsal area, TD, and the temporal posterior area, TP, exist along the rostrolateral border of V2 in tree shrews; suggest visual involvement of at least three other areas, the temporal inferior area, TI, the temporal anterior lateral area, and the temporal posterior inferior area located more ventrally in the temporal cortex; and fortify the conclusion that TD is the likely homologue of the middle temporal visual area of primates. Because tree shrews are considered close relatives of primates, the evidence for several visual areas along the border of V2 is more compatible with theories that propose a series of visual areas along V2 in primates, rather than a single visual area, V3. *J. Comp. Neurol.* 401:109–128, 1998.

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Tree shrews have long been considered of great interest for comparative studies of the visual system, both because they are highly visual mammals (e.g., Lund et al., 1985), and because they were considered to be primitive primates (e.g., Le Gros Clark, 1959) or close relatives of primates (Campbell, 1980; Cronin and Sarich, 1980; Lockett, 1980; Martin, 1990; Simmons, 1993). Tree shrews, as the sole members of the order Scandentia, are grouped presently in the higher-level taxon, Archonta, with Chiroptera (bats), Dermoptera (gliding lemurs), and primates. Studies of the visual systems of these four related lines of evolution have the potential of revealing which features are held in common by Archontan mammals and which features are specializations of a particular group. Because of their

phylogenetic relationship, studies of tree shrews have particular value for understanding the origin and evolution of distinctive features of primate brains.

Our special interest in the present study was in using patterns of connections in conjunction with cortical architecture to develop and evaluate theories of organization of extrastriate visual cortex in tree shrews. In tree shrews,

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two visual areas, area 17 (V1) and area 18 (V2), are obvious and their features have been well described (e.g., Kaas et al., 1972). These visual areas appear to have emerged early in mammalian evolution and are retained in most or all extant mammals (see Kaas, 1995a). The connection patterns of these well-established visual areas can be used to help identify the locations and extent of additional areas of cortex involved in processing visual information. This approach has already been used in tree shrews, in which V1 has been shown to project in different densities and topographic patterns to three regions along the rostrolateral boundary of V2 (Sesma et al., 1984). A large oval of temporal cortex along the mid-portion of V2, proposed as the temporal dorsal area (TD) receives dense inputs from V1. A more posterior region, proposed as the temporal posterior area (TP) receives a dense pattern of projections from V1, while a temporal anterior region (TA) receives few inputs from V1. These three proposed visual areas differ in myeloarchitecture, with TD being most myelinated. More recently, TD and TA have been shown to express higher levels of an antigen for the Cat-301 antibody than TP (Jain et al., 1994).

In the present study, we used restricted injections of several different tracers in V1, V2, TD, and cortex even more laterally in the temporal lobe to further evaluate the proposal that at least three visual areas exist along the outer border of V2, and to determine the locations of other visual areas in the temporal cortex. We considered both ipsilateral and callosal patterns of connections. The more-distributed ipsilateral pattern was more useful in estimating the extent of visual regions, whereas the more-restricted and disjunctive callosal pattern was more suggestive of separate fields (see Kaas, 1995b). Connections of V1 and V2 were of special interest, because representations of the contralateral visual hemifield for these visual areas are well-established (Kaas et al., 1972), and topology of connections of regions of the temporal cortex with V2 suggests the locations and retinotopic organizations of areas in the temporal cortex.

There are several specific questions that we hoped to address. Extrastriate visual cortex in primates clearly consists of a large number of areas (Kaas, 1989, 1994, 1998), perhaps more than 30 in Old World monkeys (Fellman and Van Essen, 1991). Most other mammals that have been investigated also appear to have many visual areas in addition to V1 and V2 (Kaas and Krubitzer, 1991), and some of these visual areas of non-primate mammals could be homologous with visual areas in primates. Of course, the close relatives of primates are most likely to have homologous areas. Most notably, area TD of tree shrews has already been proposed as a possible homologue of the middle temporal visual area (MT) of primates because of similarities in relative location, connections with V1, myeloarchitecture, and staining with Cat-301 antibody (Sesma et al., 1984; Kaas and Preuss, 1993; Jain et al., 1994; Northcutt and Kaas, 1995). The present study of connection patterns allowed us to further evaluate this hypothesis. Another issue concerned the number of visual areas that border V2 in tree shrews. In monkeys, some investigators have proposed that the outer border of V2 is largely or completely occupied by a single visual area (V3), whereas other investigators have argued that a sequence of several visual areas form the outer border of V2 (see Kaas, 1995a, 1996, 1998 for review). Although this issue of extrastriate visual cortex organization can be addressed

by further studies in primates, the organization of extrastriate cortex in tree shrews and other close relatives of primates is certainly relevant. If tree shrews have a single visual area (V3) along the outer border of V2 then it would seem more likely that the common ancestors of tree shrews and primates had a V3, and that V3 might have been retained in some or all primates. On the other hand, if tree shrews have many visual areas along the outer V2 border, then this condition would seem more likely as an ancestral condition that has been retained in tree shrews and in some or all extant primates. Some of our results have been briefly presented elsewhere (Lyon et al., 1996, 1997).

MATERIALS AND METHODS

Patterns of connections were investigated by placing injections of one or more distinguishable tracers into visual cortex of nine adult tree shrews (*Tupaia belangeri*). All procedures were approved by the Vanderbilt Animal Care and Use Committee, and followed NIH guidelines. Tree shrews were anesthetized with ketamine hydrochloride (30 mg/kg, IM) and xylazine (10 mg/kg, IM), placed in a stereotaxic headholder, and a small portion of dorsolateral visual cortex was exposed. Injections of tracers were placed at one or more cortical sites (Table 1), with targets estimated visually using distances from the midline and the caudal pole, as well as from blood vessels that course along the lateral border of the primary visual cortex, V1 (Pritzell et al., 1988). Tracers for anatomical connections included 0.02 μ l of 2% wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP), 0.2 μ l of 2% biotinylated dextran amine (BDA, 1:1 mixture of 10,000 and 3,000 MW, Molecular Probes, Inc., Eugene, OR; Naito and Kita, 1994; Roullier et al., 1994) or one or more of several fluorochromes. The injected fluorochromes included 0.2 μ l of 2% Diamidino Yellow (DY; Keizer, et al., 1983), 10% Fluororuby (FR; Schmued et al., 1990), or 10% Fluoroemerald (FE; Novikova et al., 1997). The use of several different distinguishable tracers allowed us to obtain more data on connections from each case, and to compare results from separate injections within each case (see Stepniewska and Kaas, 1996). In addition, WGA-HRP and BDA provided information on anterograde transport. BDA was especially useful in revealing the details of local connection patterns. Following injections, the dura was replaced over the cortex and covered with a soft, disposable contact lens to further protect the cortex. The opening in the skull was sealed with a thin cap of dental acrylic, the temporal muscle sutured across the midline, and the skin opening closed with suture. The animals were allowed to recover from anesthesia after injections of appropriate antibiotics and analgesics.

Five to seven days later, the animals were given a lethal dose of sodium pentobarbital and, when areflexive, perfused transcardially with 0.9% saline followed by a solution of 2% paraformaldehyde fixative, and subsequently 2% paraformaldehyde with 10% sucrose in phosphate buffered saline (PBS). The brains were removed from the skull, and the cortex was separated from the brainstem and carefully flattened and held in place between glass plates, which were submerged in 30% sucrose in PBS overnight. The flattened hemispheres were then frozen and cut parallel to the pial surface into 40- μ m sections. A series of every fourth section was mounted on glass slides

without further processing to observe fluorescent tracers. The remaining sections were divided into three sets that were variously processed for WGA-HRP with tetramethylbenzidine (TMB; Mesulam, 1981), BDA (immunocytochemical procedure of Roullier et al., 1994), myelin (Gallyas, 1979), the Cat-301 antigen (Hockfield and McKay, 1983; Jain et al., 1994), or cytochrome oxidase (CO; Wong-Riley, 1979).

Labeled neurons and terminations were plotted at high magnification by using a computerized system coupled to the microscope stage (Bioquant Image Analysis System, R&M Biometrics, Nashville, TN) or drawn at high magnification by using a microscope drawing tube. Cell plots were then carefully aligned with drawings of adjacent sections at the same magnification by using local landmarks such as blood vessels for reference. These drawings of adjacent brain sections contained boundaries of cortical areas determined from the myelin, Cat-301, and CO patterns.

Photomicrographic images (Figs. 2, 3, 5, 6) were obtained using a Leaf Lumina digital scanning camera mounted on a Nikon E800 microscope. Each digitized image was adjusted as a whole for brightness and contrast using Adobe Photoshop, but not altered in any other way.

In the present study, injection sites and distributions of labeled cells and terminals were related to subdivisions of the cortex that were architectonically identified or estimated by position relative to identified areas. Figure 1 shows some of the cortical areas of interest on a dorsolateral view of the brain. The proposed visual areas include V1 and V2 and the bordering TA, TD, and TP (see Sesma et al., 1984). More temporal areas or regions proposed here are the temporal inferior area (TI), the temporal anterior lateral area (TAL), and the temporal posterior inferior area (TPI). More rostrally, a somatosensory region includes the first and second somatosensory areas, S1 and S2 (shown as S; Sur et al., 1981a,b; Weller et al., 1987). Rostrolaterally, an auditory region largely or solely consists of the first auditory region, A (Casseday et al., 1976; Oliver et al., 1976; Oliver and Hall, 1978).

RESULTS

The results are divided into several sections. First, we present an overview of the cortical areas of the tree shrew brain (Fig. 1) and how they were identified architectonically. Second, we briefly present results from injections in V1, because these connections have been previously reported. Third, we consider our results from injections placed at various locations along the rostrocaudal extent of V2. Fourth, we describe the distributions of transported label after injections in the region of TD along the lateral border of V2. Finally, we present results from injections placed even more laterally in the temporal lobe in areas identified as visual by the above experiments.

Architectonic subdivisions of cortex in tree shrews

Several subdivisions of cortex can be reliably identified in brain sections from the experimental tree shrews cut parallel to the surface and processed for myelin, CO, or Cat-301. While brain sections cut in the coronal and parasagittal planes are extremely useful, the full extent of

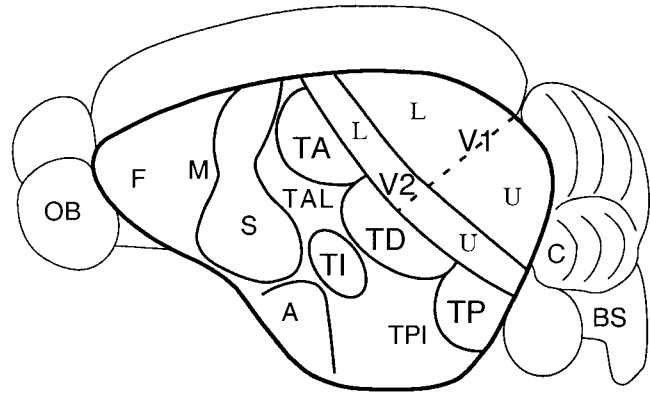


Fig. 1. Proposed cortical areas in tree shrews shown on an outline of the dorsolateral view of the brain. Visual areas V1 and V2 are based on Kaas et al. (1972). In both of these fields, the upper visual quadrant (U) is represented caudally, and the lower quadrant (L) rostrally. The dashed line represents the horizontal meridian of V1 and V2. The V1/V2 border represents the vertical meridian for both V1 and V2. The temporal anterior (TA), temporal dorsal (TD), and temporal posterior (TP) areas are outlined along V2, and the myelin-dense temporal inferior (TI) area is shown. Locations of other proposed areas, the temporal posterior inferior (TPI) and temporal anterior lateral (TAL) areas are shown relative to somatosensory (S), auditory (A), frontal (F), and motor (M) cortical regions. OB, olfactory bulb; C, cerebellum; BS, brainstem. Medial is towards the top and rostral is to the left.

areas can be most readily appreciated in sections cut parallel to the cortical surface, as in the present cases. In Figure 2, for example, V1 is most obvious in the caudomedial portion of a section stained for myelin. Because the flattening of cortex was uneven, the section crosses many cortical layers and sublayers and staining is uneven. Thus, the darkest part of V1 corresponds to darkly staining middle layers of cortex, whereas the lighter outer rim corresponds to more superficial parts of layer 3. Within these darker and lighter regions, the patchy local variation in staining density (Fig. 3) may relate to the modular organization of V1 (Fitzpatrick, 1996). Despite these variations within V1, the lateral border of the area is obvious as a sharp line of transition to the less myelinated V2.

V2 is apparent as a moderately myelinated 1.5-mm wide band of cortex along the lateral border of V1 (Cusick et al., 1985). In Figure 2, the more caudal portion of V2 is less dark and less obvious because this portion of the section includes more superficial layers. By examining several sections, the full extent of V2 can be determined with assurance. Within V2, patches of darker myelin with lighter surrounds are often apparent. The lighter surrounds have the densest interhemispheric connections (Cusick et al., 1985).

More laterally, areas TA, TD, and TP differ slightly in myelination, with TD being somewhat darker. However, these slight differences were not always apparent in our experimental material, and thus we also estimated borders based on proposed locations along V2 (Sesma et al., 1984; Jain et al., 1994). The regions or areas we denote as TAL and TPI were not distinguished architectonically. However, the proposed area, TI, was obvious as a more densely myelinated oval in temporal cortex (Fig. 2). Somatosensory cortex was also easy to identify as a more densely myelinated region. Likewise, the auditory region was consistently more myelinated than the surrounding cortex.

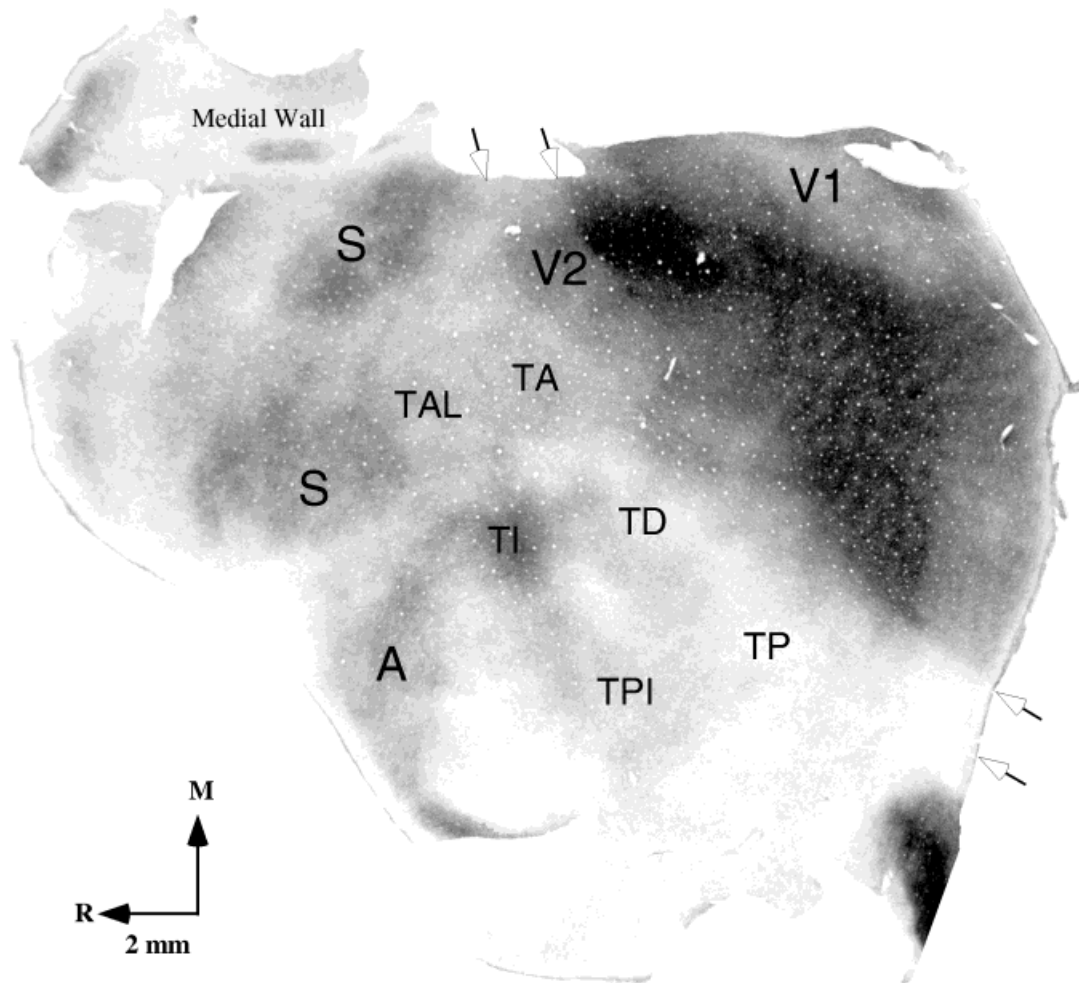


Fig. 2. A brightfield photomicrograph of a section of flattened cortex cut parallel to the surface and stained for myelin (case 96-66). The lateral border of primary visual cortex, V1, is obvious due to a sharp decrease in myelination from V1 to V2. The staining in V1 is uneven because of imperfections in flattening. The darkest staining in V1 is in the middle cortical layers (seen here in the rostrolateral region), whereas the mediocaudal fringe through upper layers is less dark. Nevertheless, a patchiness is apparent throughout V1 (see also

Fig. 3). Open arrows indicate the borders of the moderately myelinated second visual area, V2. The outer boundary of V2 can be seen due to a further decrease in myelination in the more rostral regions (it is especially clear rostrally in the section shown). Other myelin dark areas include the conspicuous temporal inferior area (TI), the somatosensory (S) region, and the auditory region (A). TD, TA, and TP, as well as the TAL and TPI regions are marked, but they were not reliably differentiated in our preparations. For abbreviations see Figure 1.

The CO and Cat-301 preparations were also useful in identifying these subdivisions of cortex, but less so in surface-view preparations. In brief, the myelin-dark regions were also dark in CO, and they expressed more of the Cat-301 antigen. Using all three preparations and sections across the thickness of cortex, V1, V2, TI, auditory, and somatosensory areas were reliably identified and delimited in the experimental material.

Connections of V1

Although we placed injections of tracers in V1 of four tree shrews, these results are only briefly described here as they are similar to previously reported findings (Rockland et al., 1982; Sesma et al., 1984). However, the present observations do extend previous conclusions, in part because previous findings depended on horseradish peroxidase (HRP) or WGA-HRP injections and the present

results include connections revealed with injections of fluorescent tracers and BDA. The results shown in Figure 4A after an injection of DY into central V1 were typical. The injections in V1 led to three main findings. First, connections were widespread and patchy within V1. Although the patchy nature was apparent in the distribution of retrogradely labeled neurons by DY, patches were more obvious when terminals were also labeled with a bidirectional tracer. A photomicrograph of such mixed label is shown after a BDA injection in Figure 5.

Patches of label surrounding injection sites extended over much of V1, but they were most dense near the injection site, where they often merged to form short bands. Second, V2 was the most densely labeled extrastriate area. Single injections in V1 labeled much of V2 (Fig. 4) but a topographic pattern was apparent across cases. Rostral V1 injections tended to label rostral V2, and caudal V1 injections tended to label caudal V2. In addition, the

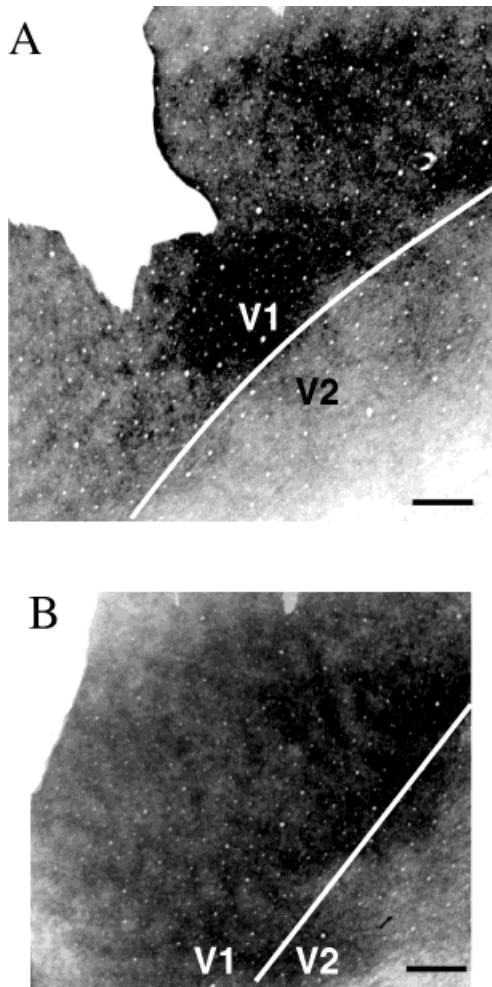


Fig. 3. The myelin-dark and myelin-light patches in the middle layers of area 17 (V1) of tree shrews 96–81 (A) and 97–29 (B). Medial is to the top and rostral is to the right. Scale bars = 1 mm.

labeled regions in V2 were patchy. The patchy pattern of V2 connections was most obvious in the anterograde pattern revealed by bidirectional tracers. Thus, several patches of label were apparent in V2 after a BDA injection in V1 (case 96–56; Fig. 5; see also DY reconstruction, Fig. 4). Third, dense connections were typically revealed within the TD region. In contrast, V1 injections produced only moderate amounts of label in TP and sparse amounts of label in TA (Fig. 4). The fluorochromes typically labeled a few more distant cells in and around TI as well.

Injections in V1 also labeled many neurons in the contralateral hemisphere (Fig. 4B), even when the injection was well within V1 (see Fig. 4A). The densest foci of callosally labeled neurons did not match the locations of the injection site. Note that with an injection centered nearly 2 mm from the V1 border, most of the labeled neurons in the other hemisphere were along the V1/V2 border. The labeled neurons corresponded to a midportion of V1 that roughly matches that of the injection site, but most of the callosally labeled neurons were more lateral than the position matching the injection site. In this case, label in V2 was most dense near the V1 border, but other injections nearer the V1/V2 border labeled more of the

width of contralateral V2. Injections in V1 labeled only a few cells lateral to V2, and these were largely in TD.

Connections of V2

Injections of tracers were placed in various rostrocaudal locations in V2 of tree shrews (Figs. 6–10). Results were fairly consistent across cases in that individual injections revealed 1) intrinsic connections that included approximately one-third the length of V2; 2) ipsilateral connections that included V1, temporal cortex immediately lateral to V2, and sometimes more lateral cortex; and 3) the contralateral connections that were most dense in the portion of V2 matching the injection site, but callosally transported label also included adjoining parts of V1 and the temporal cortex. In addition, distributions of ipsilateral label varied in V1 and to a lesser extent in the temporal cortex with the location of the injection site in V2. Many of these features of the connection pattern of V2 were apparent in case 96–81, in which a small injection of BDA was placed near the rostrolateral border of middle V2 (Fig. 6A, 7). Despite the small size of the injection, labeled cells and terminals (Fig. 6A) were distributed along 5 to 6 mm of the length of V2. A similar patchy pattern of label in V2 was apparent after a BDA injection on the rostral border of V2 that extended into TD (Fig. 6B). As a result of the V2 injection in case 96–81, a major focus of label in V1 was in the caudal half and displaced from the lateral border (Fig. 7), a location roughly corresponding in visual hemifield representation to about 10° from the vertical meridian, just above the zero horizontal meridian (Kaas et al., 1972). The visuotopic location of this focus corresponds closely with that of the injection site in V2 (Kaas et al., 1972). A few scattered cells were labeled in the portion of V1 between this focus and the injection site. In addition, foci of labeled cells were found over a broad expanse of temporal cortex, including areas TA, TD, and TP. Several foci of label were in the TD region. Contralaterally, labeled neurons were found in the matching part of V2, and adjoining V1 with a few cells in TD (not shown).

A similar but more expansive pattern of transported label was seen after an injection of FR in central V2 (case 96–12, Fig. 8). Again, labeled neurons were scattered over much of V2, most within a 6- to 7-mm strip centered on the injection site. However, a few labeled cells were even more distant. Likewise in V1, most labeled neurons were clustered laterally near the injection site in V2, reflecting a visuotopic matching of connections between the two areas, but other labeled cells were scattered over much of the rostral half of V1. There were also many labeled cells scattered over more anterior and ventral parts of temporal cortex in proposed areas TAL, TI, and TPI, and in the cortex of the medial wall. Contralaterally, labeled neurons were concentrated in the part of V2 matching the injection site and especially in the adjoining part of TD and possibly TA. A cluster of labeled cells was in lateral V1, while a few labeled cells were scattered in anterior and inferior temporal cortex, within proposed areas TAL and TI. The more scattered pattern of transported label after the FR injection was characteristic of the results after injections of fluorescent tracers, whereas more focused patches were seen after BDA or WGA-HRP injections.

In case 96–43 (Fig. 9A), an injection of WGA-HRP was placed in a rostral V2 location that closely corresponded to the FR injection in case 96–12 (Fig. 8). The injection core of WGA-HRP was larger than that of the FR injection, and it

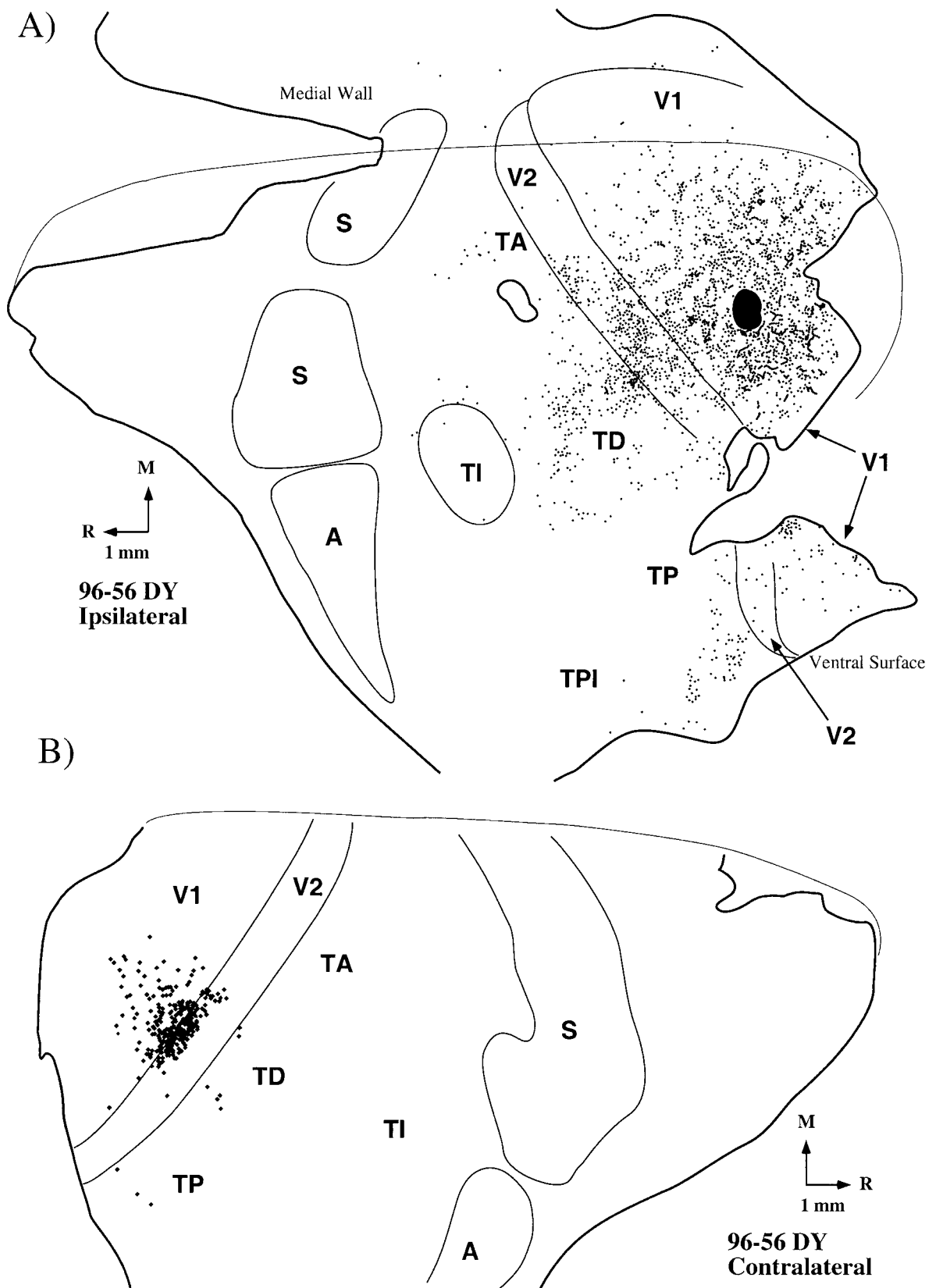


Fig. 4. Distributions of transported label in the (A) ipsilateral and (B) contralateral hemispheres after an injection of Diamidino Yellow (DY) into V1. Black oval marks the injection site and dots mark labeled neurons. Note the extensive and patchy distribution of labeled

neurons within ipsilateral V1, and the location of contralaterally labeled neurons on the V1/V2 border. Thin lines mark the border of surface cortex for ease of comparison with Figure 1. The abbreviations used for cortical areas are as in Figure 1.

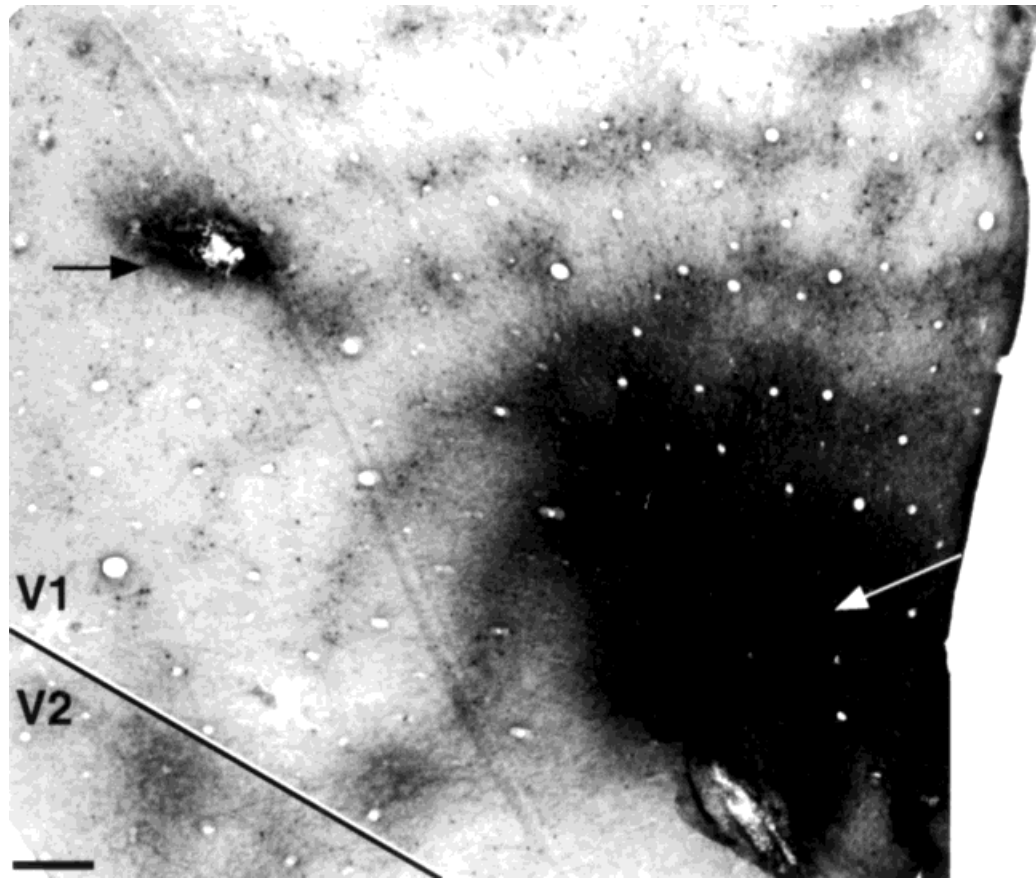


Fig. 5. A photomicrograph showing the distribution of label in area 17 (V1) and area 18 (V2) after an injection of biotinylated dextran amine (BDA) into V1 for case 96-56. Note the widespread distribution of labeled neurons and terminals in bands and patches around the

injection site. A patchy pattern is also apparent in V2. The white arrow shows the BDA injection site, and the black arrow marks the location of the DY injection illustrated in Figure 4. Medial is towards the top and rostral is to the left. Scale bar = 250 μ m.

slightly included margins of V1 and temporal cortex lateral to V2. Nevertheless, the basic pattern of labeled regions closely corresponded to that of the smaller FR injection. A comparable zone of adjacent V1 was most densely labeled. Intrinsic connections were again most dense within several millimeters of the injection core, although they also extended over approximately one-third of the length of V2. The TA, TD, and TP regions were labeled, with a concentration of label again in ventral TD. Scattered cells were located in more anterior and ventral temporal cortex. Again, contralateral label in V2 was dense in the location that matched that of the injection site, with less label in adjacent V1 and adjacent temporal cortex (Fig. 9B). The WGA-HRP injection did not label neurons ipsilaterally on the medial wall, as extensively over V1, or as densely in contralateral temporal cortex as for the FR injection in case 96-12.

The results from the injection in rostral V2 of case 96-43 are of further interest because an injection of DY was placed in caudal V2 in the same hemisphere (Fig. 9C). Thus, the results of rostral and caudal V2 injections could be compared in the same case. The DY injection labeled many V2 neurons within 2 to 3 mm of the injection site, a broad region of adjoining V1, and temporal cortex, especially the TP region adjacent to the injection site. The TD

region near V2 was also labeled, and a few scattered cells were labeled in ventral temporal cortex. Contralaterally, neurons were labeled in caudal V2, the bordering V1, and the bordering TP-TD junction (Fig. 9D).

Results from two other cases (Fig. 10) further extend these observations on how connection patterns vary with location of the injections in V2. FE was injected near the rostral pole of V2 in case 96-66 (Fig. 10A), FR was injected at the caudal pole of V2 in case 96-81 (Fig. 10B). As expected, label was most dense in portions of V2 near the injection core, adjoining parts of V1, and adjoining parts of the temporal cortex. Contralateral foci of label (not shown) were also in rostral or caudal poles of V2 for the FE or FR injections, respectively, and in adjoining parts of V1 and temporal cortex. Quite significantly, the injections also produced patches of labeled neurons in the ipsilateral temporal cortex that were not adjacent to the injection site. Thus, the rostral and caudal V2 injections produced foci in rostral and caudal TD, respectively. However, the rostral injection did not label neurons in TP, and the caudal injection did not label neurons in TA. Finally, the rostral injection labeled a scattering of neurons in more anterior and ventral temporal cortex, extending into the TAL and TI regions.

Injections in the temporal belt (areas TA, TD, and TP)

Injections of one or more tracers were placed in cortex lateral to V2 in five tree shrews. The results confirm those from V1 and V2 injections in showing that a dorsal band of temporal cortex is interconnected with both of these fields. The results also demonstrate interconnections between fields in temporal cortex, and indicate by connections that more ventral portions of temporal cortex are also involved in visual processing. As for V2, callosal connections of these areas are most dense at homotopic positions.

After an injection of FR in the TD portion of the dorsal temporal band of cortex, labeled neurons were densely packed within 2 mm of the injection site (Fig. 11). These labeled neurons may largely reflect intrinsic connections within TD. Other labeled neurons were located more caudally in TP, whereas only a few scattered neurons were located more rostrally in TA or more ventrally in temporal cortex. However, many labeled neurons were located in the middle of the V2 band and in the middle of V1, portions of the fields devoted to frontal vision near the zero horizontal meridian. Labeled neurons in the contralateral hemisphere were concentrated in TD and the midportion of V2, with fewer labeled neurons in TA and V1. An injection of FE in a slightly more rostroventral location in TD produced quite similar results (Fig. 12). Again the transported label was most dense within roughly 2 mm of the injection site, other label was concentrated in the midportions of V1 and V2, and some labeled neurons were in TA and TP. The major difference in the two cases was scatter of labeled neurons over more anterior and ventral regions of temporal cortex after the more lateral TD injection (Fig. 12) including the TAL and TI regions. Labeled neurons were also scattered over more of V1. In the contralateral hemisphere, as for case 96-43, a dense focus of labeled neurons was in TD in a location matching the injection site, and a few labeled neurons were in TA, the midportion of V2, and even V1.

Several other injections involved the TD region. In case 96-66, FR was injected rostrally near the TA and TD border, possibly largely in TA (Fig. 13A). FR label was found in V1, but distinct puffs of labeled neurons were found in midcaudal V2. Most of the labeled neurons in the temporal cortex were scattered across the TD region, with a dense focus in area TP. In the same case, BDA injections in medial TD that extended slightly into V2 produced scattered foci of labeled neurons and terminals across the middle half of V2, nearby V1, and over much of the TD region (Fig. 13B). In case 97-18, FE was injected in caudal TD near the TP junction, possibly including area TP (Fig. 13C). Labeled neurons were densely scattered over rostral portions of TD and TP, and less densely scattered in TA and more ventral regions of temporal cortex, including area TI. The label in V2 and V1 was much more caudal than in cases of more rostral TD injections. Similar, but more widespread connections were revealed in a second case with a caudal TD injection (97-29; not shown). Finally, a caudoventral TD injection, possibly extending outside the TD border, labeled neurons in caudal V1 and V2, many neurons near the injection site, and neurons scattered over much of temporal cortex, including the TI and TPI regions (Fig. 14). Contralaterally, labeled neuron largely matched the location of the injection site.

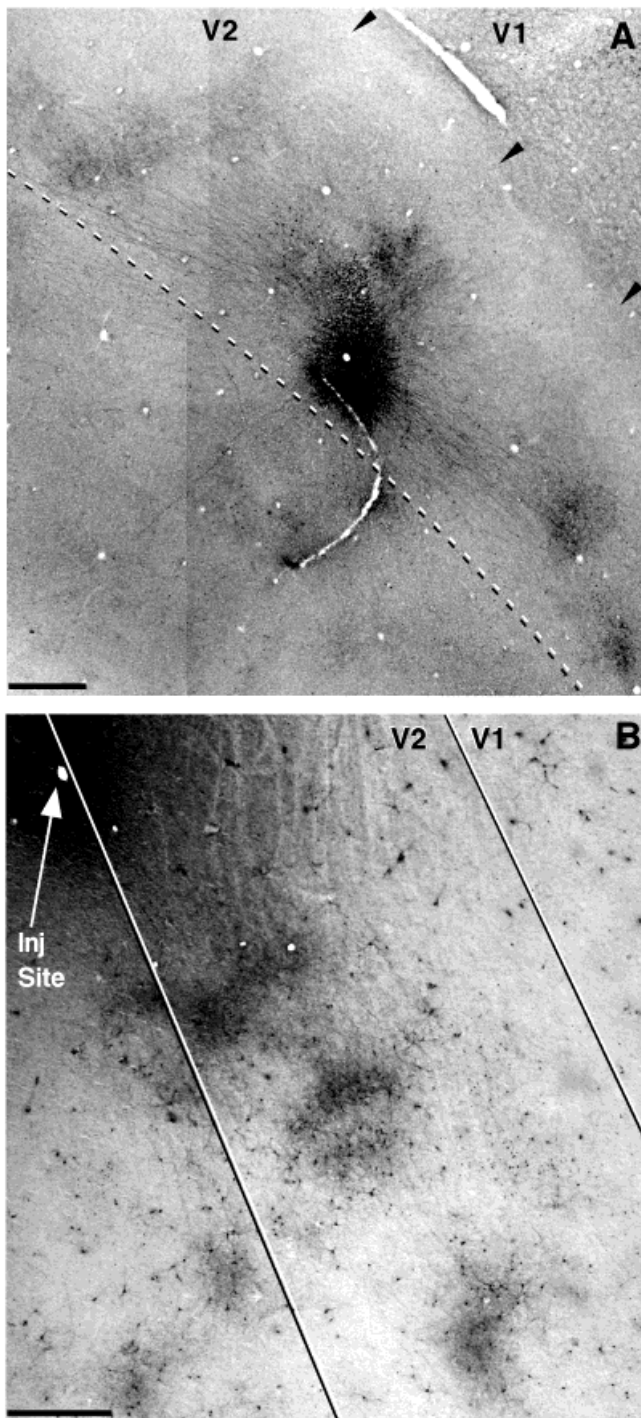


Fig. 6. Photomicrographs of transported label after injections of biotinylated dextran amine (BDA) into (A) V2 of tree shrew 96-81 and (B) at the V2/ TD border of tree shrew 96-66. The injections labeled clusters of cells and terminals over approximately one-third of the length of V2. Note the patchiness in the distribution of labeled cells. The injection site in A is the dark oval near the lateral V2 border. The white arrow in B indicates the injection site. Arrowheads in A indicate the V1/V2 border. The black/white lines indicate the lateral V2 border in A and B as well as the V1/V2 border in B. Medial is towards the top and rostral is to the left. Scale bars = 500 μ m.

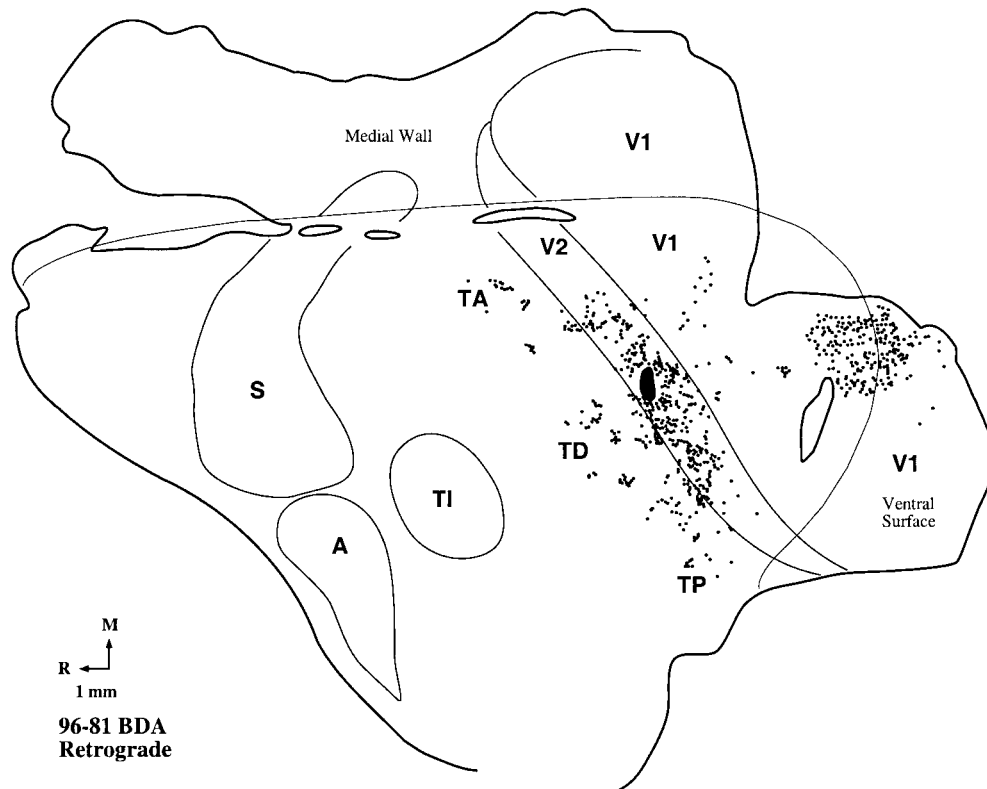


Fig. 7. Transported label after an injection of biotinylated dextran amine (BDA) into lateral V2 in tree shrew 96-81. The labeled cells shown here were coextensive with labeled terminals (not shown). Conventions as for Figure 4.

DISCUSSION

In the present study, we placed injections of distinguishable tracers into V1, V2, and the adjoining temporal cortex and examined the resulting clusters of labeled cells in order to reveal patterns of cortical connections relative to established cortical architecture. Our goal was to further determine how visual cortex is subdivided into areas and interconnected in tree shrews. The results fortify and extend earlier observations on the intrinsic ipsilateral and contralateral connections of V1, and provide the first observations on connections revealed by injections confined to V2 and to the temporal dorsal region. Here, we review this evidence in relation to the most relevant previous findings on the connections of visual cortex in tree shrews, and discuss the implications of the results in the broader context of the evolution of visual cortex organization in Archontan mammals, including primates.

Connections and modules of V1

Injections of tracers in V1 revealed widespread intrinsic connections, ipsilateral connections with V2, TD, and sometimes TP and TA (Fig. 15). Callosal connections were with V1, V2, and TD. The widespread intrinsic pattern of connections of V1 has been previously described (Rockland and Lund, 1982; Rockland et al., 1982; Sesma et al., 1984). The descriptions across these studies were highly similar in that horizontal connections spread 2 to 3 mm or more

from the margins of injection sites, thereby interconnecting many widely separated neurons without overlapping receptive fields. The only notable difference in reports was that Rockland and coworkers emphasized a banding pattern of intrinsic connections, while Sesma and coworkers focused on the patchiness of the pattern. The slight differences in illustrated results is perhaps related to the use of different tracers, HRP or WGA-HRP. The present similar results are again based on different tracers, BDA and fluorescent dyes. These methods reveal patches of labeled neurons and terminals that tend to bead together to form bands near the injection site. A reasonable interpretation of these connections, based on injections into neurons of known orientation selectivity relative to the overall pattern of orientation selectivity in V1 (Fitzpatrick, 1996; Bosking et al., 1997), is that they connect neurons across visual space with similar orientation preferences. This interpretation explains much of the pattern, but not necessarily all of the patchiness. Another factor may be that a separate modular pattern is superimposed on the parallel-band/pinwheel-like pattern of orientation-specific domains. Primates have a modular pattern within V1 of blobs and surrounds, with the blob subunits tending to connect more extensively with other blobs (e.g., Cusick and Kaas, 1988). Tree shrews do not have blobs (e.g., Wong-Riley and Norton, 1988), but our myelin-stained sections demonstrate an uneven distribution of myelin-dark and myelin-light patches in flattened preparations (Figs. 2, 3). Thus, there is architectonic evidence for a tiling of two

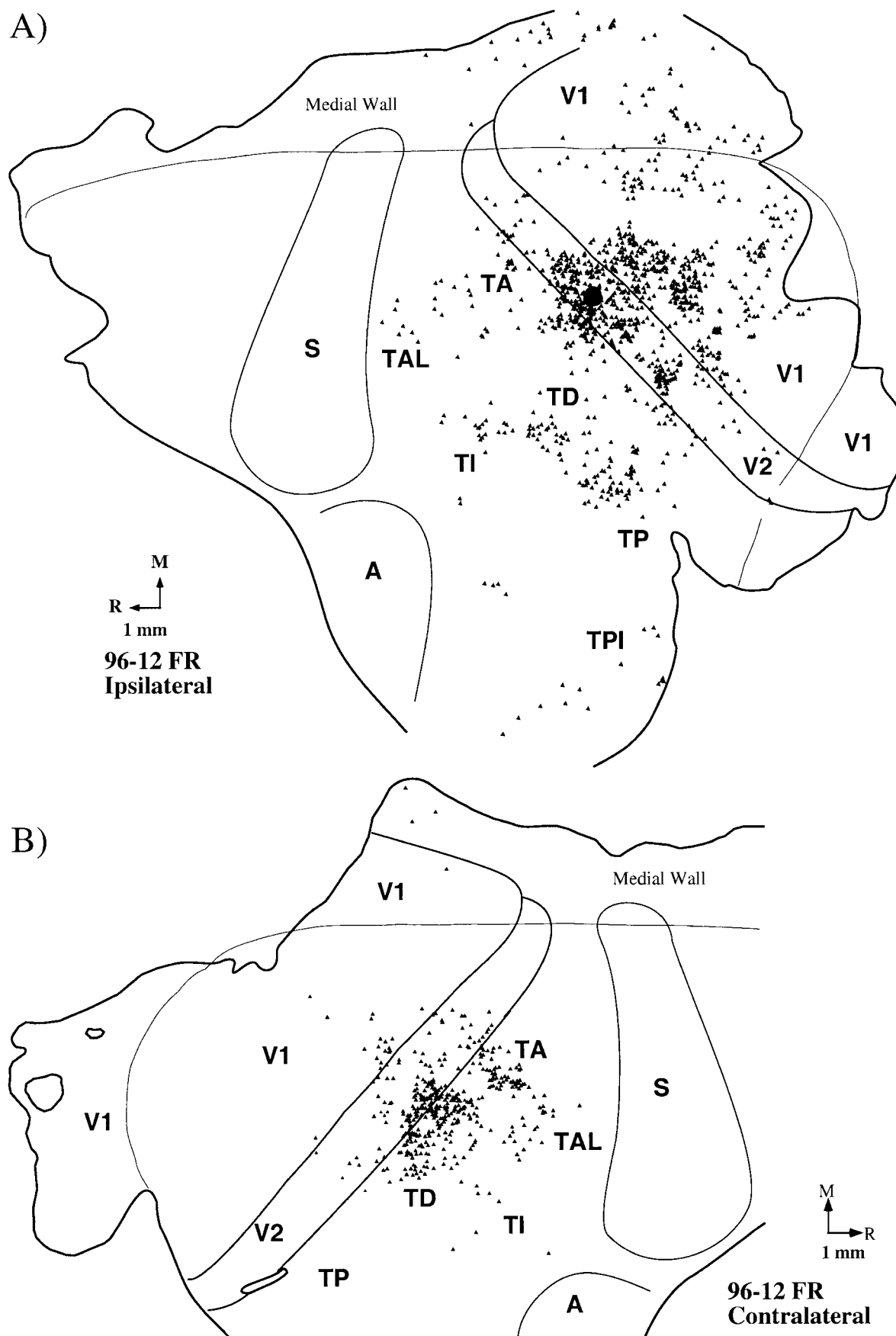


Fig. 8. Distribution of labeled cells in the (A) ipsilateral and (B) contralateral hemisphere after an injection of Fluororuby (FR) in V2 of tree shrew 96-12. Conventions as for Figure 4.

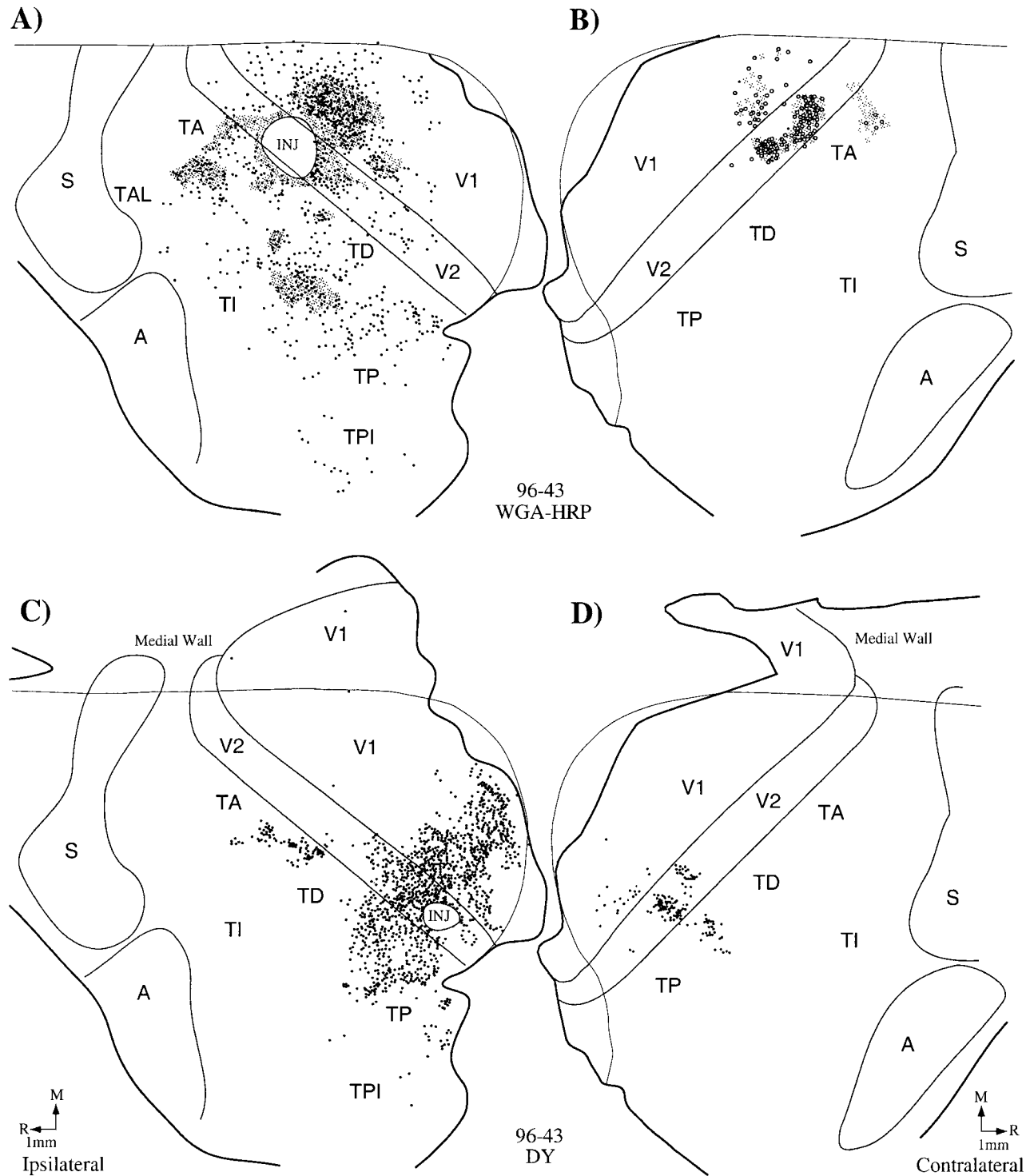


Fig. 9. Transported label after an injection of wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP) in rostral V2 (A,B), and DY in caudal V2 (C,D) of tree shrew 96-43. A and C are the injected ipsilateral hemispheres and B and D show the label in the

contralateral hemisphere, respectively. INJ, injection site. Dense stippling indicates labeled terminals; dots, labeled neurons. Other conventions as for Figure 4.

types of patch-like modules of unknown significance in V1 of tree shrews. A modular pattern related to other aspects of visual processing, superimposed on the orientation

selective bands, could produce the observed patchy pattern of intrinsic connections, which are generally thought to connect groups of neurons with similar response proper-

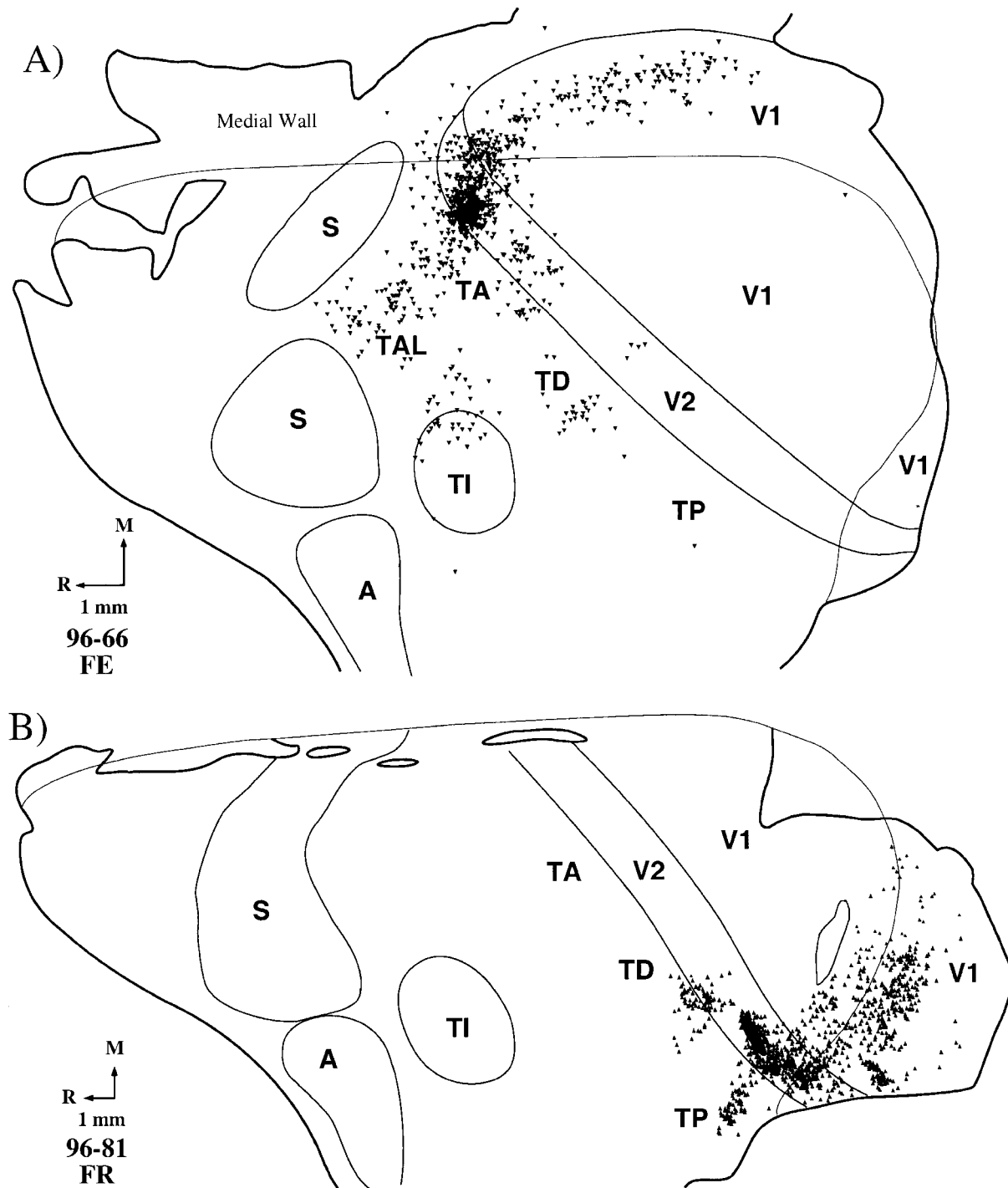


Fig. 10. Transported label after injections of fluorescent tracers at opposite poles of V2. **A:** Fluoroemerald (FE) injection in rostral V2 of tree shrew 96-66. **B:** FR injection in caudal V2 of tree shrew 96-81. Conventions as for Figure 4.

ties (e.g., Calloway, 1998; Singer and Gray, 1995; see also Kisvárdy et al., 1997).

Connections and modules of V2

The present results provide the first description of connections of V2 in tree shrews based on injections

confined to V2. As for the intrinsic connections of V1, the intrinsic connections in tree shrew V2 are extensive and patchy. Small injections in V2 label neurons over approximately one third or more of its length, thereby connecting neurons of 15° or more of visual field representation and many neurons with receptive fields that completely fail to

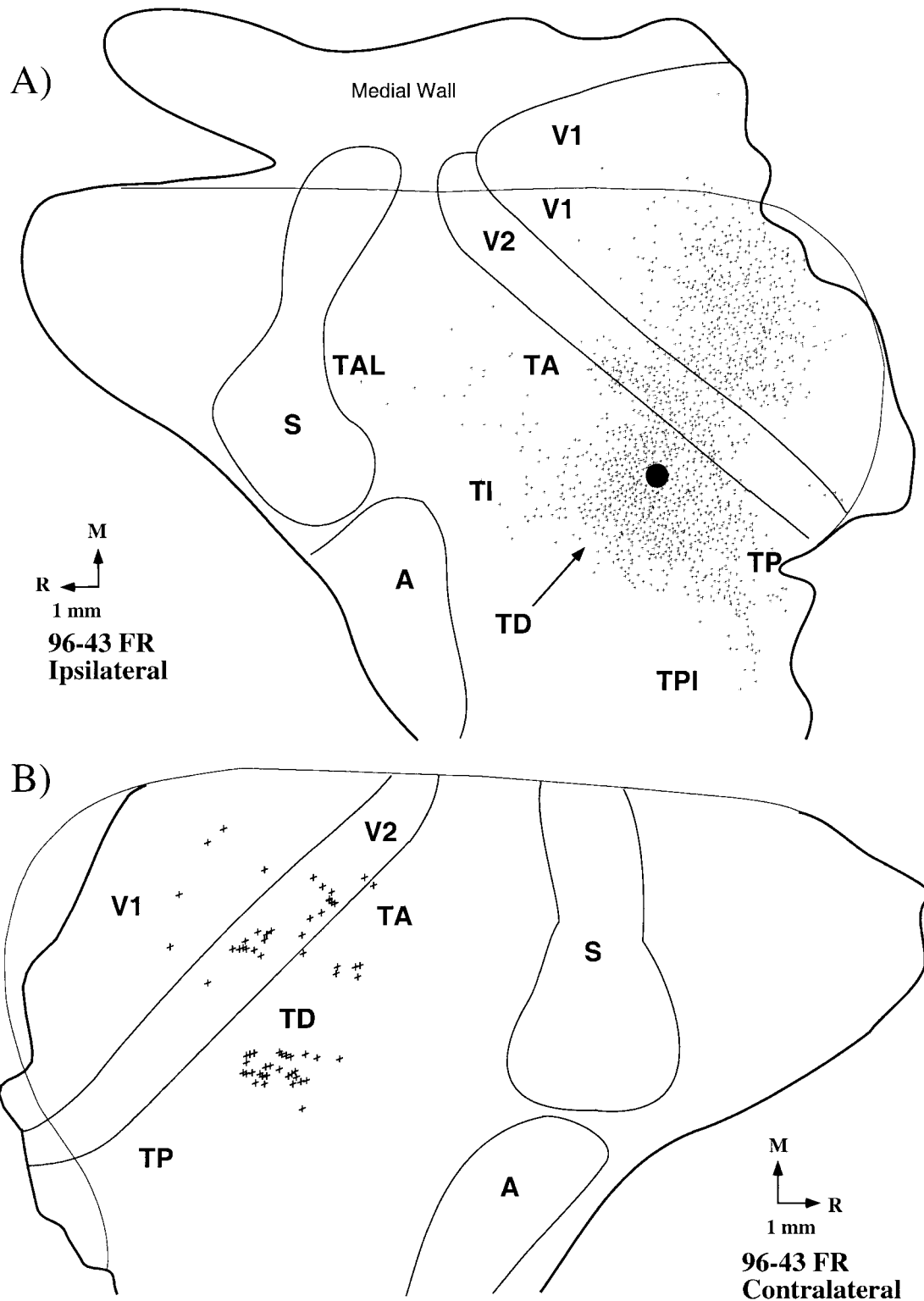


Fig. 11. Transported label after an injection of FR in the TD region of tree shrew 96-43. **A:** Injected hemisphere. **B:** Contralateral hemisphere. Conventions as for Figure 4.

over-lap. Such a spread of intrinsic connections might be expected, because intrinsic connections spreading over 10 mm or more of V2 have been reported in macaque monkeys

(e.g., Levitt et al., 1994; Kennedy and Bullier, 1985; Rockland, 1985; Stepniewska and Kaas, 1996). The intrinsic connections in tree shrews appeared to be quite patchy

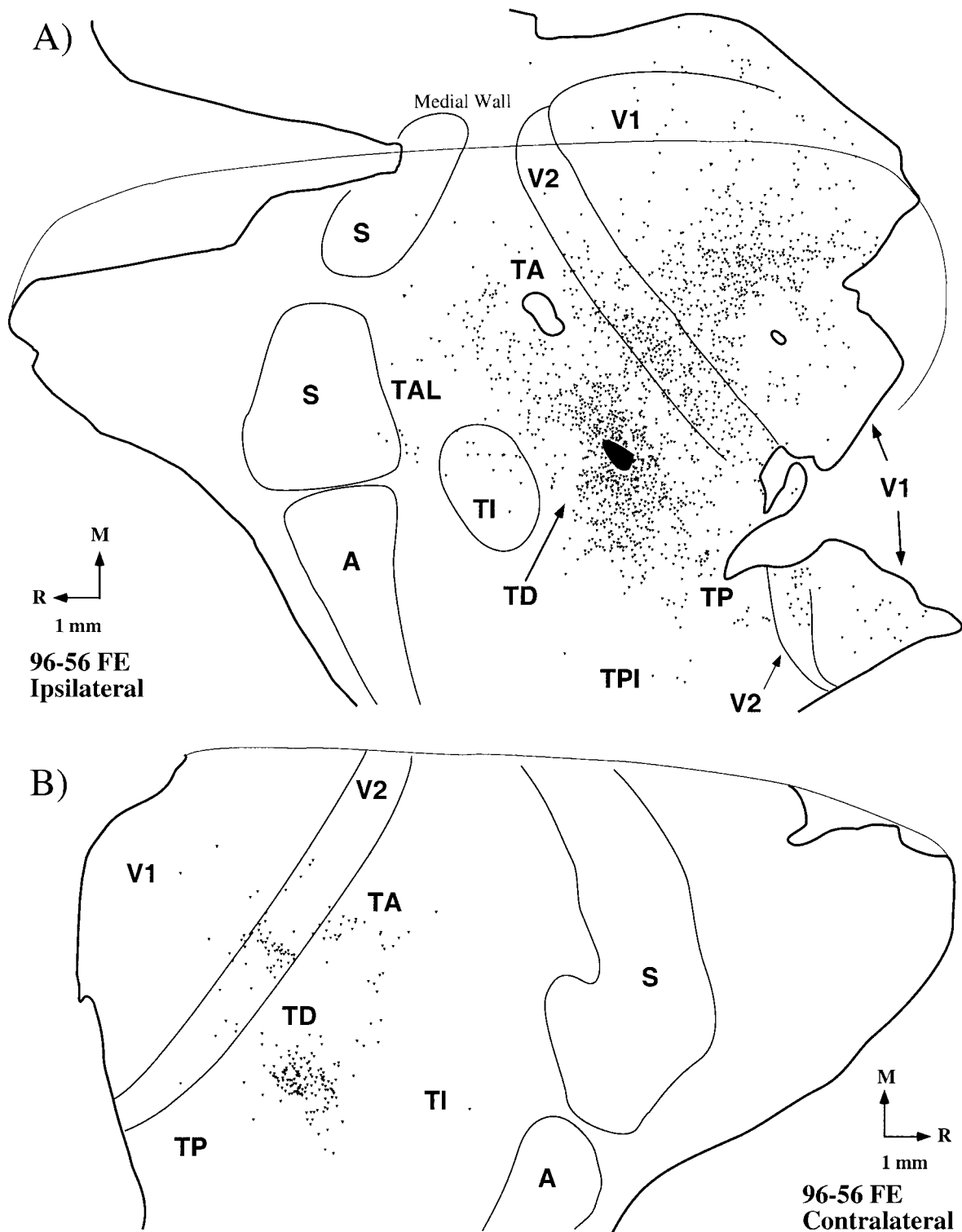


Fig. 12. Transported label after an injection of Fluoroemerald (FE) in the TD region of tree shrew 96-56. **A:** Injected hemisphere. **B:** Contralateral hemisphere. Conventions as in Figure 4.

in distribution, especially after the confined BDA injections, with patches centered 0.5 to 1.0 mm from each other. This, of course, suggests that some sort of modular organization exists in V2 of tree shrews. We expect connections to

be concentrated between groups of neurons with similar response properties, while avoiding zones of neurons with contrasting response properties (e.g., Calloway, 1998). For example, at least some of the three types of band-like

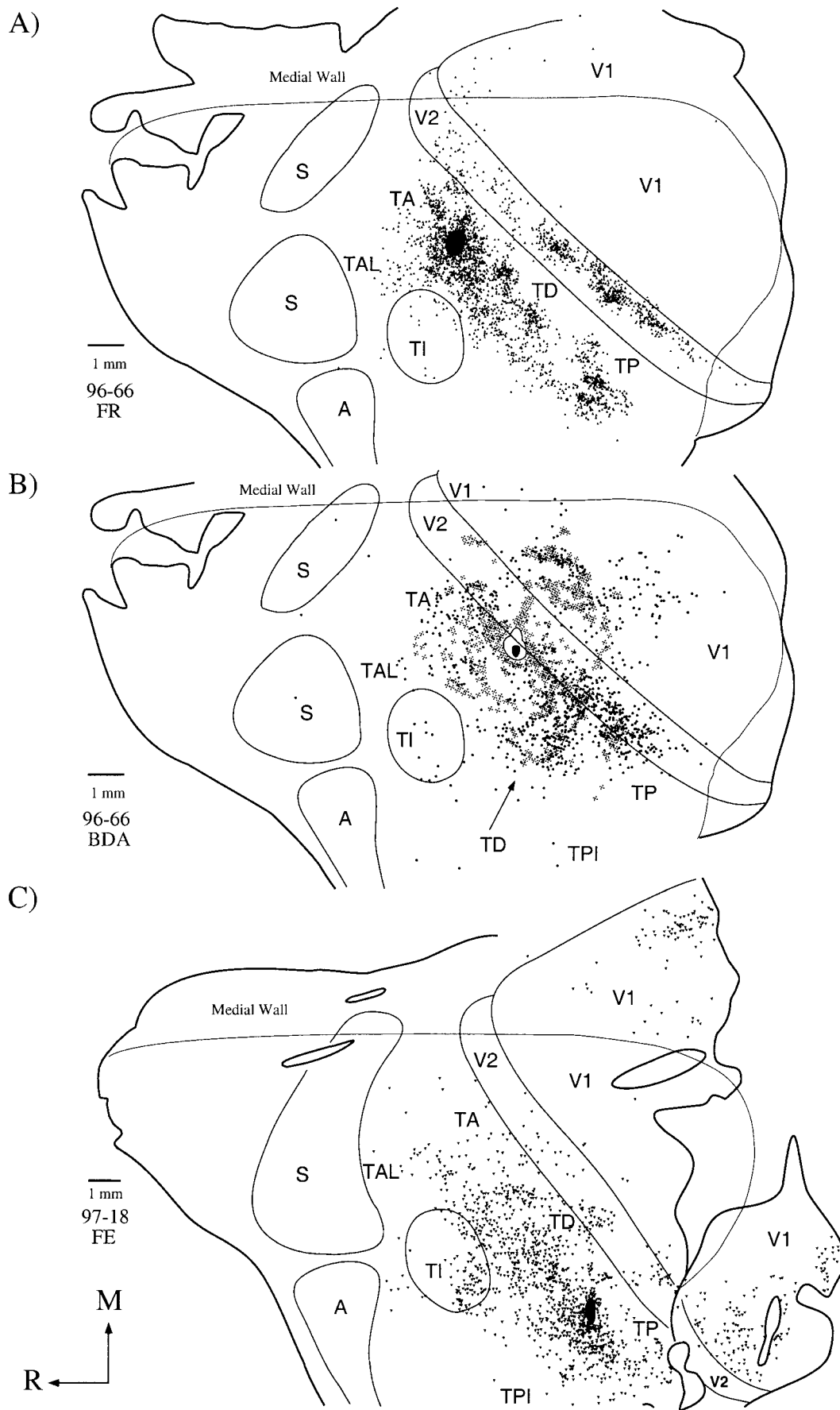


Fig. 13. Transported label after injections in three different locations in the temporal belt. **A:** An injection of FR at a location that includes both the caudal TA and rostral TD areas. **B:** An injection of BDA in medial TD. A small part of the injection site involved lateral V2. **C:** A FE injection in caudal TD and rostral TP. Conventions as for Figure 4.

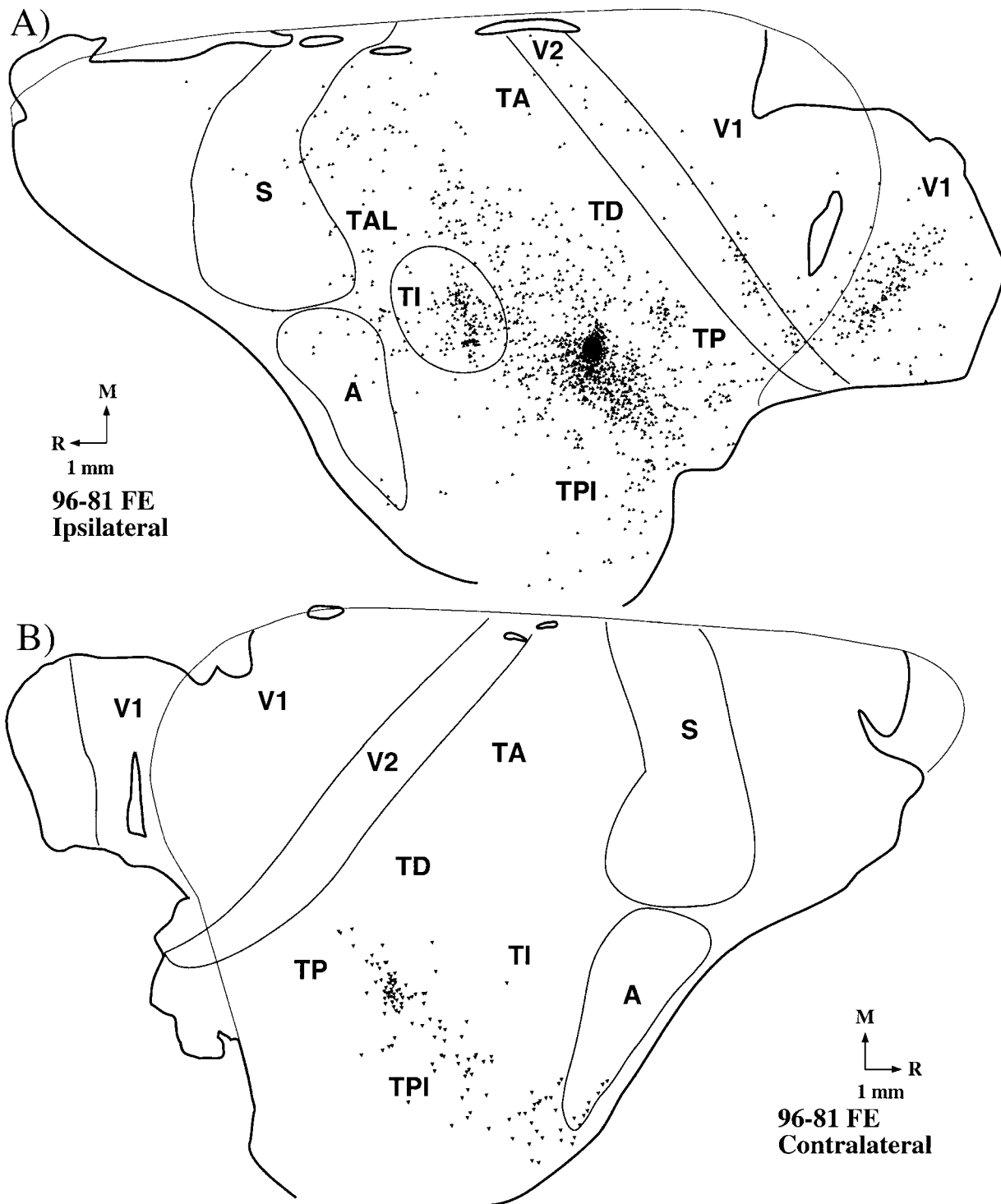


Fig. 14. Transported label after an injection of FE in the region of lateral TD and TP areas. **A:** Injected hemisphere. **B:** Contralateral hemisphere. Conventions as for Figure 4.

modules that subdivide V2 of monkeys are preferentially interconnected with each other (Levitt et al., 1994), and these intrinsic connections also appear to favor neurons of similar orientation selectivity (Malach et al., 1994). Although we do not know from electrophysiological studies

how V2 in tree shrews might be subdivided, there is additional anatomical evidence for modular subdivisions. First, optical imaging studies suggest a weak pattern of orientation selective bands in V2 (Bosking et al., 1997). More notably, inputs from V1 are distributed in a patchy

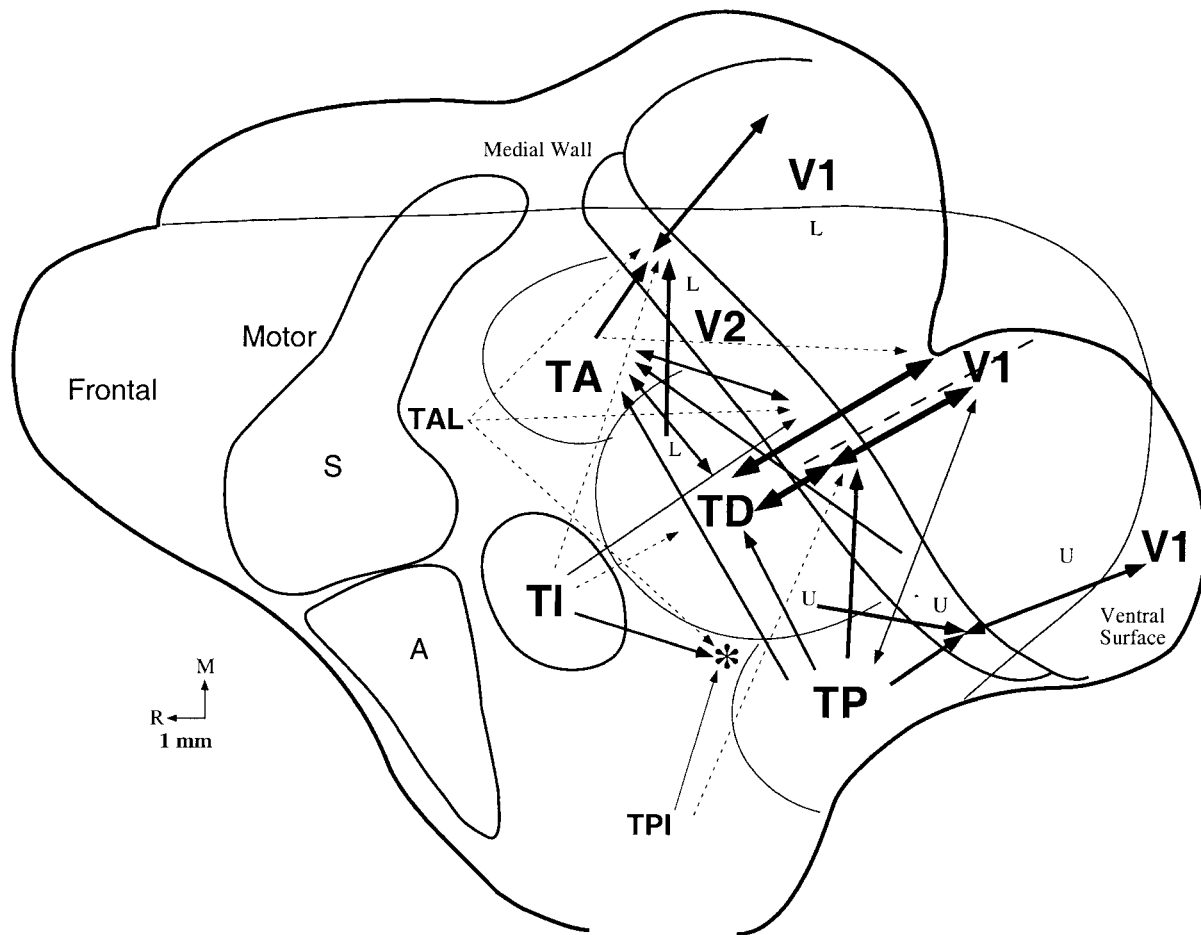


Fig. 15. Summary of connections of tree shrew visual cortex. Double-headed arrows indicate reciprocal connections between areas. Thicker lines indicate a stronger projection, dashed arrows indicate

the weakest connections. The asterisk near the rostral end of the TP border represents the general location of connections that could not be reliably assigned to either TD or TP. For abbreviations see Figure 1.

pattern (present results; Sesma et al., 1984), and callosal connections are concentrated in repeating band-like zones that separate and merge to surround 0.5-mm ovals within V2 that have few callosal connections (Cusick et al., 1985). The callosal-dense zones are also myelin-dense, whereas callosal-sparse zones are myelin-light. However, we do not yet know how patterns of striate cortex inputs, callosal connections, and intrinsic connections of V2 relate to each other. A similar modular pattern of myelin-light ovals and myelin-dense surrounds exists in the well-differentiated V2 of squirrels (Kaas et al., 1989).

As expected, injections of tracers in V2 consistently labeled neurons in V1. In all studied mammals, V1 and V2 are interconnected (see Kaas and Krubitzer, 1991 for review). In addition, the connection pattern was largely between retinotopically matched locations in the two representations in that caudal V2 injections labeled caudal V1 and rostral V2 injections labeled rostral V1. Nevertheless, there was considerable spread in the distributions of neurons projecting to V2, with the dense core of labeled neurons in the retinotopically appropriate location in V1 surrounded by scattering of labeled neurons over adjoining cortex. Possibly some of the neurons were labeled artifactually by tracer leaking from injection sites, since there was

considerable variability in the distribution of the sparsely scattered neurons labeled by V2 injections. Yet, these sparsely distributed labeled cells may reflect the full extent of the convergence of connections from V1 to V2. In cats, projections over approximately 7 mm in the rostrocaudal direction of V1 converge to a single location in V2 (Henry et al., 1991), and similar values of convergence have been obtained in monkeys (Parker et al., 1986). When the proportions of V1 and V2 are considered rather than absolute sizes, the extent of the densely labeled zones in V1 after V2 injections in tree shrews appear to be roughly comparable to those in cats and monkeys, but the sparse distributions of labeled cells in V1 are proportionately more extensive. Many of the inputs to V2 from the densely labeled zones in V1 would contribute to the approximately 4- to 8-times larger excitatory receptive fields of neurons in V2 (Kaas et al., 1972), but others may mediate effects beyond the classical receptive field (Allman et al., 1985).

The lateral connections of V2 largely were with a broad belt of adjoining cortex containing proposed areas TA, TD, and TP. In addition, a scattering of labeled cells was located more laterally in temporal cortex in a pattern that was quite variable across injections. The connection pat-

tern provided evidence that each area, TA, TD, and TP, is interconnected in a patchy fashion with V2. In addition, the results suggest that more rostral injections in V2 tend to label more rostral sites in TD, and more caudal injections, more caudal parts of TD. Although most injections in V2 labeled portions of TA and TP as well as TD, the most caudal injections (Fig. 10B) failed to label neurons in TA, whereas the most rostral injection (Fig. 10A) failed to label neurons in TP. It is possible that our short survival times were inadequate to reveal these farthest connections. These two cases, however, did provide some of the more compelling evidence that rostral V2, representing the lower quadrant, projects to rostral TD, and caudal V2, representing the upper quadrant, projects to caudal TD. The patches of label in TD were the most consistent features of the total pattern, and they often contained the most label.

Connections of V2 with cortex lateral to V2 have been reported for a number of mammalian species (see Kaas and Krubitzer, 1991), and this may be a feature common to all mammals with V2. The pattern seen in tree shrews resembles that found in squirrels in which at least two visual areas with V2 connections appear to border V2 (Kaas et al., 1989). In prosimian primates and monkeys, V2 projects to several bordering areas, including the dorsomedial area, DM (or dorsal V3 by some interpretations), and the caudal dorsolateral area (DLc) or caudal V4 (e.g., Cusick and Kaas, 1988; Stepniewska and Kaas, 1996; Pospichal and Kaas, 1996). In cats, the connection pattern of V2 has been interpreted quite differently as to a single adjoining area, V3, and the data seem quite consistent with this interpretation (e.g., Pospichal et al., 1996). Thus, the organization of cortex just lateral to V2 seems to be variable across species, but the evidence in squirrels, megabats, tree shrews, and primates suggests that V2 projects to two or more adjoining areas in these mammals.

Callosal connections for cortical areas in general are most dense between the same fields, but they also include two or more adjoining areas (see Cusick and Kaas, 1988). The pattern of callosal connections of V2 in tree shrews fits this expectation. Injections most densely labeled neurons in contralateral V2, more sparsely in V1, and sometimes in or near TD. The focus of labeled neurons in contralateral V2 matched the rostrocaudal level of the injection site, and they spread across the width of V2. A similar pattern of callosal connections for V2 has been described for squirrels (Kaas et al., 1989). In monkeys, callosal connections of V2 include V2, DLc, and V1 (Tigges et al., 1974; Kennedy et al., 1986; Cusick and Kaas, 1988; see Kaas, 1995b for review), with connections being most dense in V2. However, callosal connections are not evenly distributed across V2, but are concentrated near the V1/V2 border. Nevertheless, they extend in finger-like bands toward the outer border of V2 within CO-dense bands (e.g., Cusick and Kaas, 1988; Olavarria and Abel, unpublished observations).

Connections of the TD region

Our injection in the TD region of tree shrews revealed widespread connections within TD, connections with temporal regions TA, and TP, dense connections with V1 and V2, and sparse and variable connections with more ventral regions of temporal cortex (Fig. 15). Callosal connections were largely with TD, but also with V2 and very sparsely V1, TP, and TA. The cortical connections of TD have not

been directly investigated before, but thalamic degeneration in the pulvinar complex has been described after lesions of the TD region (Diamond et al., 1970) providing further evidence that TD is involved in processing visual information. In addition, we know from previous results (Sesma et al., 1984) that TD is a major projection zone of V1.

The present results provide compelling evidence that TD is densely interconnected with both V1 and V2. Injections in V1 and V2 consistently labeled the TD region, and injections in TD consistently labeled V1 and V2. In addition, there was some evidence for a topographic pattern of interconnections. Injections in central TD, whether along the V2 border or near the ventral border of TD, labeled midportions of V1 and V2 (Figs. 12A, 13B). The most rostral injection labeled a scattering of cells in the rostral tips of V1 and V2 (Fig. 13A). The denser foci of label over the middle to caudal portions of V2, of course, do not fit this proposed pattern, but they could reflect the involvement of TA in the injection. The most caudal injection in TD labeled caudal V1 and caudal to middle V2. Other evidence for this crude pattern of rostrocaudal matching of connections comes from locations of label in TD after injections in V1 and V2 (present results; Sesma et al., 1984).

The injections in TD also labeled neurons in the TA and TP regions. These regions also have connections with V1 and V2, although these were more variable and sparse than for TD. Connections with more anterior and ventral parts of the temporal lobe were unexpectedly sparse, with labeled neurons consistently in the TI region, and a scattering of neurons in the TAL and TPI regions. Thus, TI, TAL, and TPI are likely to have roles in visual processing, but these regions may not be predominantly visual in function. None of the injections in TD labeled neurons in the frontal cortex, although such connections are common from higher order visual areas in primates (e.g., Bullier et al., 1996).

Implications of results for theories of visual processing in tree shrews and the evolution of visual cortex organization in primates

The present results support the conclusion that visual cortex in tree shrews includes the two major divisions of V1 and V2, as in other mammals, as well as a series of probably three visual areas along the lateral border of V2, with the likely involvement of at least three distinct regions of more temporal cortex in visual processing. Thus, evidence from the connection patterns implicates much of the temporal cortex of tree shrews in visual function. This conclusion is highly consistent with much earlier evidence that lesions of temporal cortex in tree shrews are followed by severe impairments in visual learning and discrimination (Killackey et al., 1971; Snyder and Diamond, 1968). Neurons in temporal cortex of anesthetized tree shrews are not very responsive to visual stimuli, but neurons in the general location of TD have been described as responding to visual stimuli while having receptive fields that were large, diffuse, and difficult to delineate (Kaufman and Somjen, 1979). However, the responsiveness of these neurons to visual stimuli was nearly abolished by V1 lesions.

The organization of extrastriate cortex in tree shrews is much like that of squirrels (Kaas et al., 1989), which are highly visual rodents. Squirrels, tree shrews, and perhaps bats, have expanded temporal lobes, as in primates, that are largely devoted to vision. In flying fox, a megachiro-

TABLE 1. Summary of Injection Sites¹

Case #	InjSite	Tracer	Figure
95-95	V1	WGA-HRP	N.S.
	V2	FR, DY	N.S.
96-12	V1	WGA-HRP	N.S.
	V2	FR	8
96-43	V2	DY	N.S.
		WGA-HRP	9A, 9B
	TD	DY	9C, 9D
		FR	11
96-56	V1	DY	4
		BDA	5
96-66	TD	FE	12
	V2	FE	10A
	V2/TD	BDA	6B, 13B
	TD/TA	FR	13A
96-81	V2	BDA	6A, 7
		FR	10B
97-18	TD/TP	FE	14
	V1	FB	N.S.
97-29	TD/TP	FE	13C
	TD	FE	N.S.
	TD/TP	FR	N.S.

¹InjSite, injection site; V1, area 17; V2, area 18; TD, temporal dorsal area; TA, temporal anterior area; TP, temporal posterior area; WGA-HRP, wheat germ agglutinin-horseradish peroxidase; FR, fluororuby; DY, diaminidino yellow; BDA, biotinylated dextran amine; FE, fluoroemerald; FB, fast Blue; N.S., not shown.

teran bat, V1 injections reveal at least two to three visual areas along the lateral V2 border (Krubitzer and Calford, 1990). This is a feature shared also by squirrels and tree shrews. We propose that this is the generalized condition for mammals, but the comparative evidence is still too incomplete to be conclusive. In addition, the current evidence indicates that carnivores may be different. However, the evidence for several visual areas along the V2 border of tree shrews and various primates (see Kaas, 1996, 1998) argues against the early view that the outer border of V2 in macaque monkeys is formed by a single area, V3 (e.g., Cragg, 1969; Zeki, 1969). This would require a major change from both the more generalized plan and the plan that seems to exist in most Archontan mammals.

The present results obviously relate to the issue of whether TD in tree shrews is the homologue of MT or any other visual area in primates. TD has some features of MT, most notably connections with V1 show a roughly comparable retinotopic organization, and possibly greater myelination, than surrounding areas (see Kaas and Preuss, 1993). Both MT (DeYoe et al., 1990) and TD (Jain et al., 1994) stain darkly for the Cat-301 antibody and both areas are similarly positioned across from the midportion of V1 and V2. What is clearly different is the position of TD along the outer border of V2, while MT is displaced into the temporal lobe by dorsolateral visual cortex, DL, which possibly emerged as a representation for detailed object vision in primates. The present results demonstrate that TD also resembles MT in having connections with V2, but other comparisons are less certain since we do not know if any of the other visual areas of temporal cortex are homologous in tree shrews and primates. Most likely, tree shrews do have visuomotor areas of the frontal cortex, but no connections of TD with frontal cortex were revealed. In four species of primates, MT connections with frontal cortex were a variable feature that was not demonstrated by all injections in MT, especially in prosimian galagos (Krubitzer and Kaas, 1990). Other connections of MT are with adjoining fields in the temporal lobe, medial cortex adjoining V2, and part of posterior parietal cortex. This pattern seems compatible with the pattern revealed by TD injections, but these connections are not specific enough to add significant support to the theory that TD is MT.

We have also considered the possibility that TI is a homologue of MT. TI is more densely myelinated than the surrounding cortex, and its position relative to V1, V2 and the auditory cortex is more compatible with that of MT with these fields. However, TI appears to lack V1 connections and has only sparse V2 connections. Thus, TI differs markedly from MT in these connections.

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